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Course Code: CSA1522

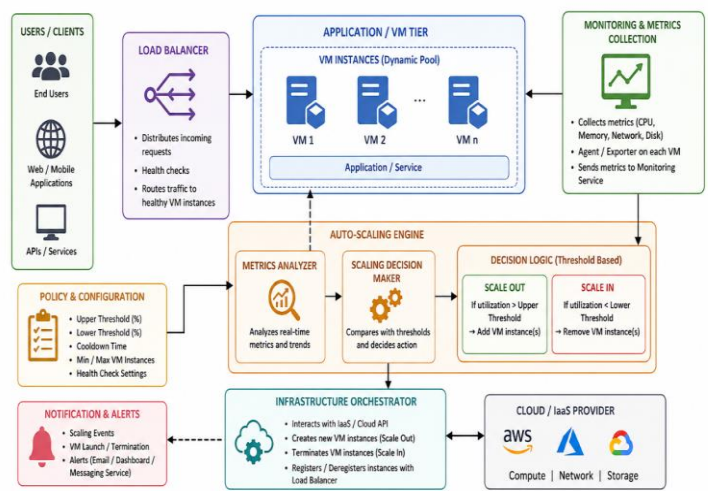
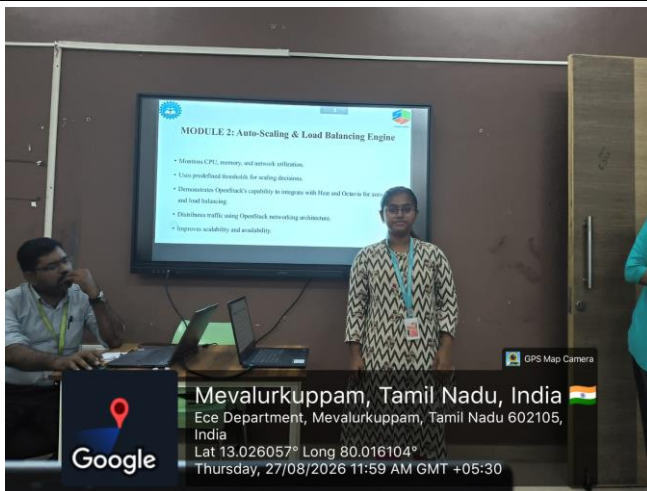
Slot: C

Course Name: Cloud Computing and Big Data Analytics

Course Faculty: Dr. P.Rajaram

Project Title: Cloud Based IaaS Virtual Infrastructure Provisioning and Auto Scaling System

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)



Project Description: (here you write what you did in this project (contribution) including Model Description)

In this module, we developed an Auto-Scaling and Load Balancing Engine that automatically manages VM instances according to changing workload conditions. The system continuously monitors CPU and resource utilization and compares the values with predefined thresholds. When the workload exceeds the upper threshold, the engine performs **scale-out** by adding new VM instances, while low utilization triggers **scale-in** by removing unnecessary instances. A load balancing mechanism distributes incoming requests across the available VMs to prevent overloading of individual instances. The module uses a **threshold-based reactive auto-scaling model**, which follows the cycle **Monitor** → **Analyze** → **Scale** → **Load Balance**, helping to maintain performance, availability, and efficient resource utilization.

Student Signature

Guide Signature